

EXPONENTIALS AND LOGARITHMS

- 1 Express each of the following in the form $\log_a b = c$.

a $10^3 = 1000$	b $3^4 = 81$	c $256 = 2^8$	d $7^0 = 1$
e $3^{-3} = \frac{1}{27}$	f $32^{-\frac{1}{5}} = \frac{1}{2}$	g $19^1 = 19$	h $216 = 36^{\frac{3}{2}}$
- 2 Express each of the following using index notation.

a $\log_5 125 = 3$	b $\log_2 16 = 4$	c $5 = \log_{10} 100\,000$	d $\log_{23} 1 = 0$
e $\frac{1}{2} = \log_9 3$	f $\lg 0.01 = -2$	g $\log_2 \frac{1}{8} = -3$	h $\log_6 6 = 1$
- 3 Without using a calculator, find the exact value of

a $\log_7 49$	b $\log_4 64$	c $\log_2 128$	d $\log_3 27$
e $\log_5 625$	f $\log_8 8$	g $\log_7 1$	h $\log_{15} \frac{1}{15}$
i $\log_3 \frac{1}{9}$	j $\lg 0.001$	k $\log_{16} 2$	l $\log_4 8$
m $\log_9 243$	n $\log_{100} 0.001$	o $\log_{25} 125$	p $\log_{27} \frac{1}{9}$
- 4 Without using a calculator, find the exact value of x in each case.

a $\log_5 25 = x$	b $\log_2 x = 6$	c $\log_x 64 = 3$	d $\lg x = -3$
e $\log_x 16 = \frac{2}{3}$	f $\log_5 1 = x$	g $\log_x 9 = 1$	h $\lg 10^{12} = x$
i $2 \log_x 7 = 1$	j $\log_4 x = 1.5$	k $\log_x 0.1 = -\frac{1}{3}$	l $3 \log_8 x + 1 = 0$
- 5 Express in the form $\log_a n$

a $\log_a 4 + \log_a 7$	b $\log_a 10 - \log_a 5$	c $2 \log_a 6$
d $\log_a 9 - \log_a \frac{1}{3}$	e $\frac{1}{2} \log_a 25 + 2 \log_a 3$	f $\log_a 48 - 3 \log_a 2 - \frac{1}{2} \log_a 9$
- 6 Express in the form $p \log_q x$

a $\log_q x^5$	b $\frac{1}{2} \log_q x^{15}$	c $\log_q \frac{1}{x}$	d $\log_q \sqrt[3]{x}$
e $4 \log_q \frac{1}{\sqrt{x}}$	f $\log_q x^2 + \log_q x^5$	g $\log_q \frac{1}{x^2} + \log_q \frac{1}{x^3}$	h $3 \log_q x^2 - \frac{1}{2} \log_q x^4$
- 7 Express in the form $\lg n$

a $\lg 5 + \lg 4$	b $\lg 12 - \lg 6$	c $3 \lg 2$	d $4 \lg 3 - \lg 9$
e $\frac{1}{2} \lg 16 - \frac{1}{5} \lg 32$	f $1 + \lg 11$	g $\lg \frac{1}{50} + 2$	h $3 - \lg 40$
- 8 Without using a calculator, evaluate

a $\log_3 54 - \log_3 2$	b $\log_5 20 + \log_5 1.25$	c $\log_2 16 + \log_3 27$
d $\log_6 24 + \log_6 9$	e $\log_3 12 - \log_3 4$	f $\log_4 18 - \log_4 9$
g $\log_9 4 + \log_9 0.25$	h $2 \lg 2 + \lg 25$	i $\frac{1}{3} \log_3 8 - \log_3 18$
j $\frac{1}{3} \log_4 64 + 2 \log_5 25$	k $\frac{1}{2} \log_5 (1\frac{9}{16}) + 2 \log_5 10$	l $\log_3 5 - 2 \log_3 6 - \log_3 (3\frac{3}{4})$